

Grid to Vehicle and Vehicle to Grid Energy Transfer using Single-Phase Bidirectional AC-DC Converter and Bidirectional DC - DC converter

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Abstract— In this paper, a configuration of a single-phase bidirectional AC-DC converter and bidirectional DC-DC converter is proposed to transfer electrical power from the grid to an electrical vehicle (EV) and from an EV to the grid while keeping improved power factor of the grid. In first stage, a 230 V 50 Hz AC supply is converted in to 380V dc using a single-phase bidirectional AC-DC converter and in the second stage, a bidirectional buck-boost dc-dc converter is used to charge and discharge the battery of the PHEV (Plug-in Hybrid Electric Vehicle). In discharging mode, it delivers energy back to the grid at 230V, 50 Hz. A battery with the charging power of 1.2 kW at 120V is used in PHEV. The buck-boost DC-DC converter is used in buck mode to charge and in a boost mode to discharge the battery. A proportional-integral (PI) controller is used to control the charging current and voltage. Simulated results validate the effectiveness of proposed algorithm and the feasibility of system.

Index Terms— Plug-in Hybrid Electric Vehicle (PHEV), Bidirectional AC-DC Converter, DC-DC Converter, Vehicle to grid (V2G), Electric drive vehicle (EDVs)

I. INTRODUCTION

The basic concept of vehicle-to-grid power is used in EDVs (Electric Drive Vehicles) to provide electric power to the grid while the vehicle is parked. The EDV can be a battery-electric vehicle, fuel cell vehicle, or a plug-in hybrid vehicle. Plug-in hybrid EDVs can function in either mode of operation. EDVs, whether powered by batteries, fuel cells, or gasoline hybrids, have within them the stored energy in the battery and power converters capable of producing a 50 Hz AC voltage that powers our homes and offices. When connections are added to allow this electricity to flow from cars to power lines, one calls it V2G (Vehicle to Grid). When connections are added to charge the battery of EDVs from power lines, then it is termed as G2V (Grid to Vehicle). PHEVs (Plug-in Hybrid Electric Vehicles) are emerging as replacement for traditional vehicles. With the energy transfer

from the grid to vehicle and vehicle to grid, the concept of smart grid is visualized. One can transfer energy from PHEV to grid when it is parked in peak hours and can take energy back from the grid in lean hours, thereby can earn some revenue as the cost of energy is much more in peak hours than in lean hours. An electric vehicle pack has a substantial amount of energy stored in its battery. One typical EDV can output over 10 kW, the average drawn power of 10 houses. The key to realizing economic value from V2G is precise timing of its grid power production to fit within driving requirements while meeting the time-critical power “dispatch” of the electric distribution system. PHEVs use grid electricity to displace transportation fuel consumption [1]. The battery plays an important role in the performance of PHEVs. Efficient charging and discharging of the battery are essential to maintain good battery life, safety, and reliability. In this paper, a configuration with the bidirectional power converters is derived for the bidirectional power management of a PHEV battery. The system can charge battery up to 10A current at 120V. It can also deliver energy back to the 230V, 50 Hz single-phase power at 10A rate. The system is composed of two parts: a single-phase bidirectional AC-DC converter and a DC-DC converter. Single-phase bidirectional AC-DC converter is to convert AC to DC voltage [1]. The buck-boost DC-DC converter is used in buck mode for charging and a boost mode when discharging the battery. The charging and discharging of the battery demonstrate the feasibility of the system. This paper is organized as follows: system configuration and principle of operation is introduced in Section-II. Design of the system is explained in Section-III. Its control algorithm is given in Section-IV, and Matlab modeling is given in Section-V. Detailed simulated results have been discussed in Section-VI. Conclusive remarks are given in Section-VII.

II. SYSTEM CONFIGURATION AND PRINCIPLE OF OPERATION

Fig. 1 shows the system configuration. A single-phase bidirectional AC-DC converter and a bidirectional buck-boost DC-DC converter are included in the system. This system includes an inductor connecting between single-phase AC and a bidirectional AC-DC converter necessary to boost dc output in order to maintain the DC bus voltage at 380V [2]. The buck-boost DC-DC converter shown in Fig.2 is used for charging in

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the buck mode and for discharging in boost mode. In charging mode, the DC-bus voltage must be higher than the battery voltage, and it operates in buck mode. Through controlling the PWM duty ratio in the buck mode, the charging current is controlled. In discharge mode, the buck-boost converter operates in the boost mode.

A. Grid-Bidirectional AC-DC Converter

In the energy transfer mode from the grid to vehicle and vehicle to the grid the interaction between the grid and bidirectional AC-DC converter is the main issue. As shown in Fig.3 during the analysis, the positive current direction is considered from the grid to an AC-DC converter. The grid voltage is considered to be a sinusoidal and it is given as,

$$v_s(t) = \sqrt{2}V_s \sin(\omega t) \quad (1)$$

where $v_s(t)$ instantaneous grid voltage with rms value of V_s . Fundamental component of an AC converter voltage is given as,

$$v_c(t) = \sqrt{2}V_c \sin(\omega t - \delta) \quad (2)$$

where V_c is rms value of fundamental component of converter voltage $v_c(t)$, and δ is an angle between $v_c(t)$ and $v_s(t)$.

The grid current is given as,

$$I_s(t) = \sqrt{2}I_s \sin(\omega t - \theta) \quad (3)$$

where θ is the angle between $i_s(t)$ and $v_c(t)$.

As illustrated in Figs. 4(a)-4(b), [3,4] an active power is provided by the grid as long as $v_c(t)$ lags $v_s(t)$, and it is sent to the grid when $v_s(t)$ lags $v_c(t)$. Since $v_c(t)$ and $v_s(t)$ are sinusoidal, $i_s(t)$ is also sinusoidal as shown before. Its phase angle, θ , determines the direction of the reactive power flow.

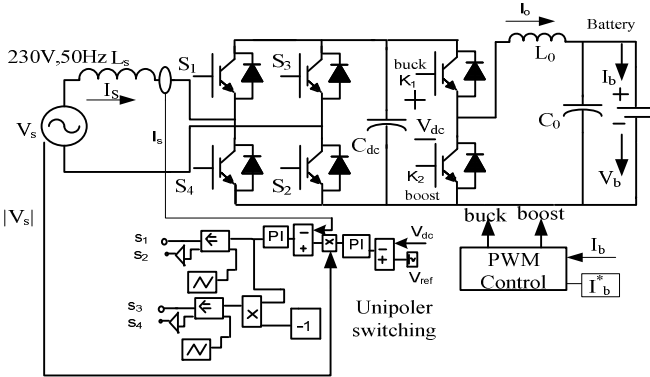


Fig.1 Proposed configuration for V2G and G2V Energy transfer

If θ is positive, the reactive power is sent to the grid, and if θ is negative, the reactive power is provided by the grid to the converter. Operating modes such as inductive and capacitive charging are shown in Figs. 4(c)-4(f) [5, 6].

III. DESIGN OF THE PROPOSED SYSTEM

The design of various components of proposed charging and discharging system consists of a single-phase bidirectional AC-DC converter, a bidirectional DC-DC boost converter, a battery energy storage system. The detailed design of each part is given in the following sections.

A. Design of Single-Phase Bidirectional AC-DC Converter

These types of converters are developed to meet the requirements of applications of bidirectional power flow in addition to improved power quality at the grid in terms of high power factor and low THD with well-regulated output DC voltage. This single-phase bidirectional ac-dc converter is designed for a 3 kW. Fig. 1 shows a circuit of this type of converter.

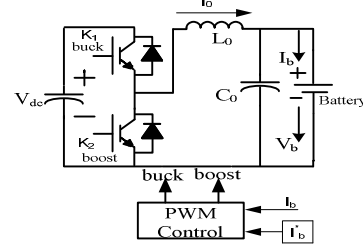


Fig.2 Bidirectional buck-boost DC-DC Converter

The fundamental converter voltage V_c is given as,

$$V_c = \frac{mV_{dc}}{\sqrt{2}} \quad (4)$$

where m is modulation index, the value of m is considered as 0.9, and V_{dc} is the dc link voltage and it is taken as 380 V. The value of V_c by using eq. (4) is 241.86V.

The relation between fundamental converter voltage and source voltage is given as,

$$V_c = \sqrt{V_s^2 + (I_s^2 * X_l^2)} \quad (5)$$

where V_s is rms grid voltage and the value of V_s is 230 V and I_s is grid rms current. By using eq.(5) the value of grid inductance is calculated as 2.1mH.

The value of dc link capacitor is given as,

$$C_{dc} = \frac{I_{dc}}{2 * \omega * v_{dcripple}} \quad (6)$$

where I_{dc} is the DC link current (P_{dc}/V_{dc}) and ω is angular frequency and $v_{dcripple}$ is 5% of V_{dc} . From eq. (6), the calculated value of C_{dc} is 1mF [7]. Detailed model parameters of the system are given in Appendix.

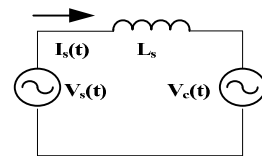


Fig.3 Representation of the grid and the charger

B. Design of Bidirectional Buck-Boost DC-DC Converter

Fig. 2 shows a bidirectional buck-boost dc-dc converter. The solid state switch K_2 is used for boosting while the switch K_1 is used for the buck mode. The relationship between switching frequency f , inductance L , in buck-boost mode is given as,

$$f = \frac{1}{2 * P * L} \left(\frac{1}{\frac{1}{V_{dc}} + \frac{1}{V_b}} \right) \quad (7)$$

Where P is conversion power, V_{dc} is input voltage and V_b is output voltage and f is the switching frequency and its value is 50 kHz. The value of P is 3 kW, V_{dc} input voltage 380V and V_b is output voltage is 120 V. From eq. (7) the value of L is

1.9 mH [8, 9]. Detailed model parameters are given in Appendix B.

C. Design of Storage Battery

A lead-acid model of the battery is implemented in simulink using model parameters given in [10]. Fig. 5 shows a Thevenin's equivalent of the battery as an energy storage unit. Its energy is represented in kWh which is stored in an equivalent capacitor (C_{bb}) expressed as,

$$C_{bb} = (kWh * 3600 * 1000) / \{0.5(V_{ocmax}^2 - V_{ocmin}^2)\} \quad (8)$$

where V_{ocmax} is the maximum voltage at the terminal of the battery when it is fully charged and V_{ocmin} is the minimum voltage at the terminal of the battery when it is fully discharged. In this Thevenin equivalent model of the battery [10], R_s is the equivalent resistance of the battery, which is usually a small value. For this analysis R_s is taken 0.01Ω. The parallel circuit of R_b and C_{bb} represents the self discharging of the battery. A typical value of R_b for this battery is considered 10kΩ. Here the battery is considered of having 1.2 kW for 12 Hr. peaking capacity, and with the variation in the voltage of order of 106 V to 136V. The calculated value of C_{bb} for this battery is from eq. (8) is calculated as 14281 F.

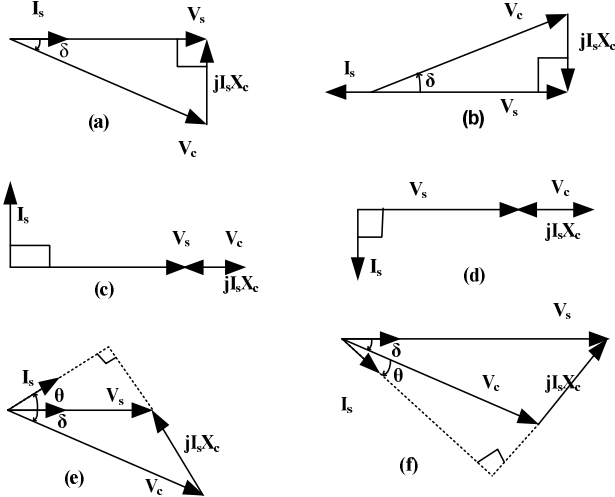


Fig.4 Vector diagram for different operating modes: (a) Charging, (b) Discharging (c), Inductive operation, (d) Capacitive operation, (e) Charging and Capacitive operation, (f) Charging and Inductive operation

IV. CONTROL ALGORITHM

The control algorithm for different blocks of proposed system is given in this section. It plays an important role in the operation of such system and is explained as follows.

A. Control of Single-Phase Bidirectional AC-DC Converter

In the control of single-phase bidirectional AC-DC converter, a unipolar switching scheme is used, in which the triangular carrier waveform is compared with two reference signals which are positive and negative signals. The output voltage varies between 0 and V_{dc} , or between 0 and $-V_{dc}$. The PI (proportional integral) voltage controller closely tracks the reference voltage (V_{ref}) and gives a control signal (I_p) to minimize the voltage error $V_e(k)$ which is calculated from the reference voltage $V_{ref}(k)$ and a sensed voltage $V_{dc}(k)$ at k_{th} instant of time as,

$$V_e(k) = V_{ref}(k) - V_{dc}(k) \quad (9)$$

The output of the controller $I_p(k)$ at k_{th} instant is given as,

$$I_p^*(k) = I_p^*(k-1) + K_{pv}\{V_e(k) - V_e(k-1)\} + K_{iv}V_e(k) \quad (10)$$

where K_{pv} and K_{iv} are the proportional and integral gains of the voltage controller.

The PI current controller closely tracks the reference current $I_p^*(k)$ and gives a control signal V_{cs} to minimize the current error $I_e(k)$ which is calculated from the reference current $I_p^*(k)$ and a sensed current $I_p(k)$ at k_{th} instant of time as,

$$I_e(k) = I_p^*(k) - I_p(k) \quad (11)$$

This current error is amplified using the proportional controller by gain "K," and which is given as,

$$V_{cs} = K I_e(k) \quad (12)$$

This amplified signal is compared with fixed-frequency (10 kHz) triangular carrier wave in unipolar PWM switching signals for the IGBTs of single-phase bidirectional AC-DC converter [7].

B. Control of Bidirectional Buck-Boost DC-DC Converter

In order to get the desired operation of charging and discharging of the battery using a bidirectional buck-boost converter, a PWM control technique is used here. A PI controller is used to control the battery output current (I_b). The PI voltage controller closely tracks the reference dc link current and gives a control signal (V_{cT}) to minimize the current error $I_{eT}(k)$ which is calculated from the reference dc link current $I_b^*(k)$ and a sensed dc link current $I_b(k)$ at k_{th} instant of time as,

$$I_{eT}(k) = I_b^*(k) - I_b(k) \quad (13)$$

The output of the PI controller $I_c(k)$ at k_{th} instant is given as,

$$V_{cT}(k) = V_{cT}(k-1) + K_{pv}\{I_{eT}(k) - I_{eT}(k-1)\} + K_{iv}I_{eT}(k) \quad (14)$$

where K_{pv} and K_{iv} are the proportional and integral gains of the voltage controller [7]. This scheme is applicable for buck as well as boost mode. The output of the controller $V_T(k)$ at k_{th} instant is compared with fixed frequency (f_s) saw-tooth carrier waveform to get the switching signals for the MOSFETs of the bidirectional buck-boost converter [7].

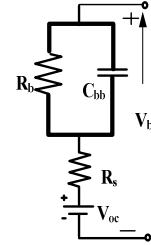


Fig.5 Thevenin's equivalent circuit of storage battery

V. MATLAB BASED MODELING

The simulation model of the proposed energy transfer from the vehicle to grid and grid to vehicle is shown in Fig.6, is developed in MATLAB. It consists of modeling of single phase bidirectional AC-DC converter. This single phase bidirectional AC-DC converter is designed for a power of a 3 kW. The bidirectional DC-DC buck-boost converter is used for charging and discharging of the battery of PHEV. The detailed parameters of bidirectional buck-boost converter are given in Appendix A. A battery energy storage system is considered for 1.20 kW for 12 Hours peaking capacity within the variation in voltage of 106 V to 136V. Detailed parameters of storage battery are given in Appendix C. Simulation is

carried out in MATLAB version of 7.7 the sim power system (SPS) toolbox using ode (23tb/stiff/TR-BDF-2) solver in discrete mode at 1e-6 step size.

VI. RESULTS AND DISCUSSION

Simulated results from the plug-in modes are shown in Figs. 7-9. The current delivered to and from the grid is shown to be sinusoidal and in phase with the grid voltage. This eliminates current harmonics and maintains a unity power

VII. CONCLUSION

The proposed converter has delivered the AC current to/and from the grid at unity power factor and at very low current harmonics which ultimately prolongs the life of the converter and the battery and minimizes the possibility of distorting the grid voltage. It also enables V2G interactions which could be utilized to improve the efficiency of the grid.

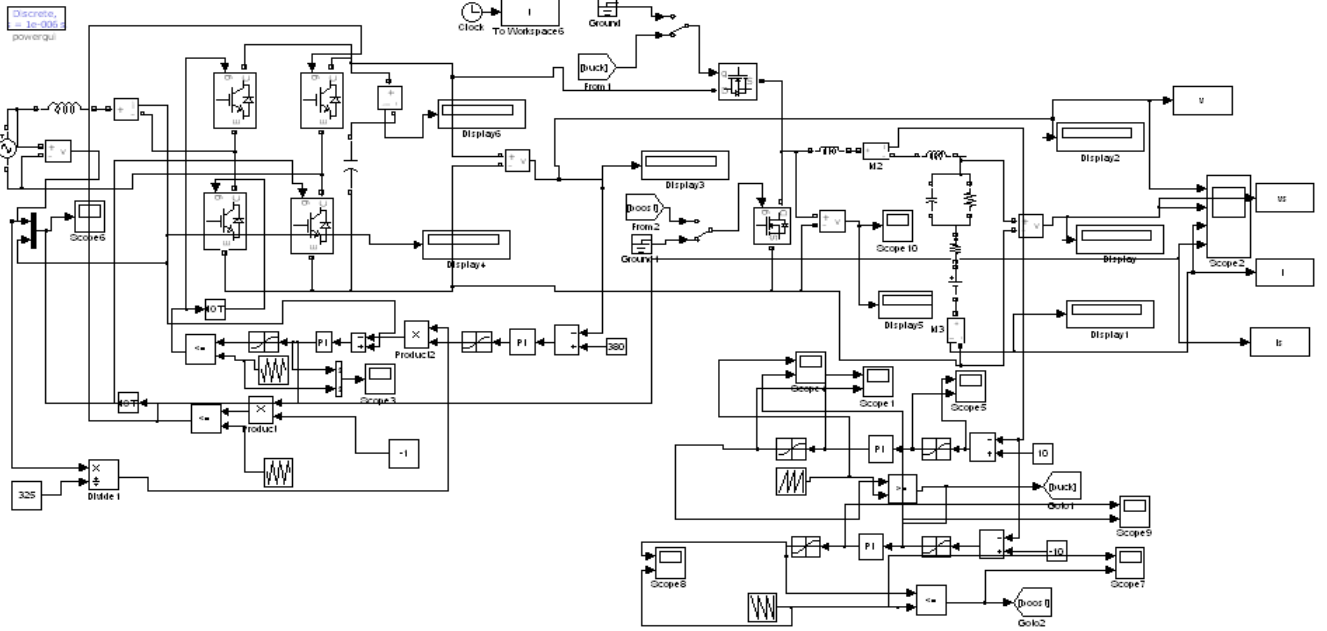


Fig.6 MATLAB/SIMULINK model for energy transfer from vehicle to grid and grid to vehicle

factor. When delivering power to the grid, the injected current is in the reverse direction of the grid voltage, which can be seen from 180° phase difference. In this case, zero crossing of the grid voltage and injected current are still matching each other. These figures show the simulated results of the loading of the DC voltage bus. Although some brief voltage transients occur during abrupt load changes, the converter maintains 380 V across the DC bus while supplying or absorbing the required current. The rise in the battery voltage while charging and fall in the battery voltage while discharging are shown in these figures corresponding to the maximum and minimum battery voltage in the charging and discharging modes, the voltage profile is demonstrated in Figs.7-8 at 1.35 to 1.45s. There is change in the mode of operation i.e. from buck mode to boost mode. In Fig.9 at 1.9 to 2.2 s. discharging to charging mode of operation is shown i.e. boost mode to buck mode and at the same point of time the direction of current is in 180° phase opposition. This shows the reversal of current and flow of power in reverse direction. In Fig.8, while showing V_s , I_s in same figure, I_s grid current is amplified by factor of 10 in order to observe it in comfortably to the given axes. Figs.10-11 shows the current harmonics spectra of charging as well as discharging grid current. The THD (Total Harmonic Distortion) of the grid current in both modes is found below a limit of 5% a limit of IEEE-519 standard.

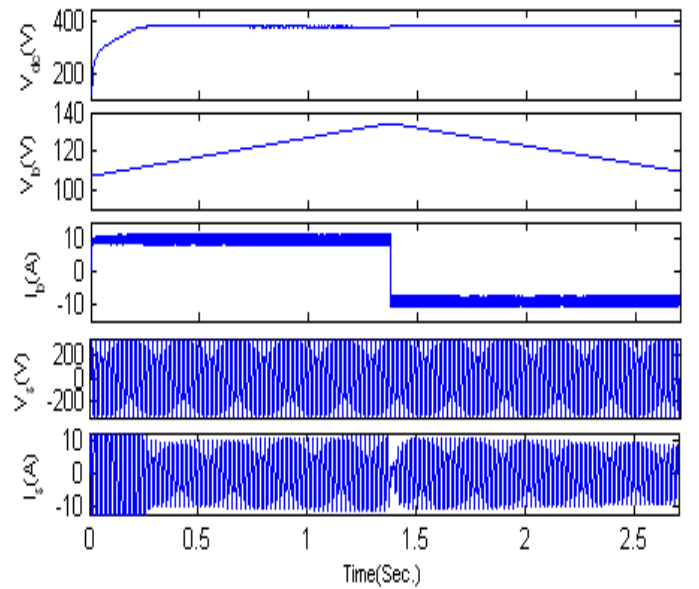


Fig.7 Charging and discharging of PHEV battery (Full profile)

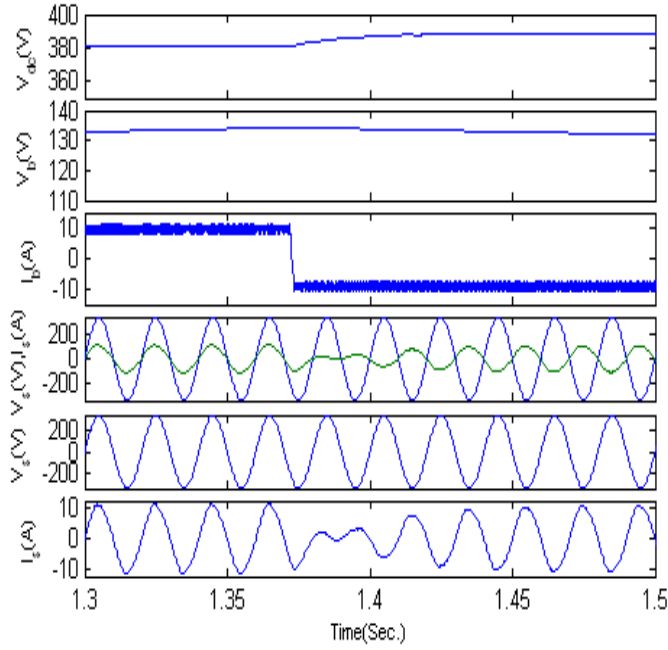


Fig.8 Charging and discharging of PHEV battery (in large view)

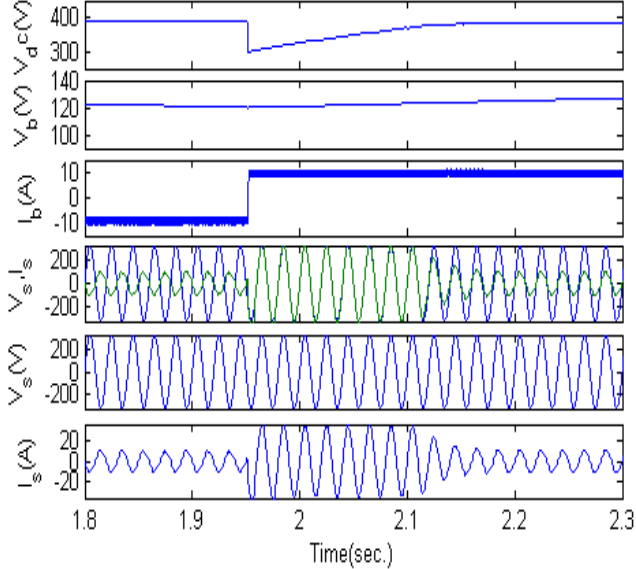


Fig.9. Discharging and Charging of PHEV battery demonstrating unity Power factor operation

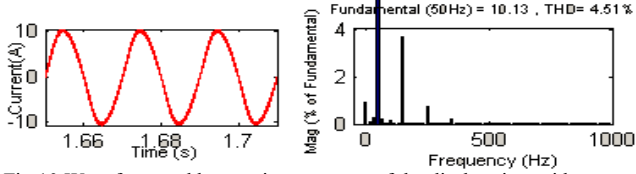


Fig.10 Waveform and harmonics spectrum of the discharging grid current

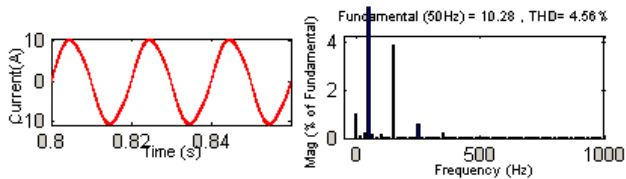


Fig.11 Waveform and harmonics spectrum of the Charging grid current

APPENDICES

A. Parameters for Single-Phase Bidirectional AC-DC Converter

$K_{i1}=2$, $k_{p1}=0.1$, $L_s=2.3\text{mH}$, $K_{i2}=2$, $k_{p2}=0.85$, 3000W , 230V rms, $f_s=20\text{ kHz}$.

B. Parameters for Bidirectional DC-DC Buck Boost Converter

Buck, $K_{i1}=1$, $k_{p1}=0.001$ for Boost $K_{i2}=0.5$, $k_{p2}=0.001$, $F_s=50\text{ kHz}$, $L_0=1.9\text{ mH}$.

C. Parameters for Storage Battery

$R_b=10\text{ k}\Omega$, $R_s=0.01\Omega$, $V_{oc}=120\text{V}$.

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